

# Anaesthesia for minimally invasive cardiac surgery

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## Learning objectives

By reading this article, you should be able to:

- Discuss the absolute and relative contraindications for minimally invasive mitral valve surgery (MIMVS).
- Explain the role of transoesophageal echocardiogram (TOE) and why it is essential in MIMVS surgery.
- Summarise the main disadvantages of a minimally invasive sternotomy compared with conventional full sternotomy for aortic valve surgery.

Significant advances in cardiac surgery followed the introduction of cardiopulmonary bypass (CPB), which was first established in the early 1950s. Over time, refinements in surgical and anaesthetic techniques combined with improved technology and the use of intraoperative transoesophageal echocardiography (TOE) has enabled less invasive approaches using smaller surgical incisions. Several different approaches can be grouped under the umbrella term ‘Minimally Invasive Cardiac Surgery’ (MICS), with Cosgrove and colleagues describing the first minimally invasive valve procedures in 1996.<sup>1</sup> Despite being pioneered more than 20 yr ago, the complex nature of MICS has resulted in only a handful of centres in the UK regularly performing these procedures. Today, MICS encompasses minimally invasive direct coronary artery bypass (MIDCAB), robotic-assisted cardiac surgery, atrial fibrillation (AF) ablation surgery, and minimally invasive approaches to the mitral valve, left and right atria, and aortic valve.

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## Key points

- The term minimally invasive cardiac surgery (MICS) comprises a variety of procedures, the most prevalent of which in the UK are port access mitral valve surgery and minimal-access aortic valve surgery.
- Not all patients are suitable; therefore, a thorough preoperative assessment and careful patient selection is essential for good outcomes.
- Intraoperative transoesophageal echocardiography plays an essential role in the safe delivery of MICS.
- Major benefits include reduced postoperative pain, shorter hospital stay, better cosmesis, and quicker resumption of normal activities.
- The anaesthetist must be aware of a range of complications specific to MICS.

Advantages of MICS over a conventional midline sternotomy include reduced postoperative pain, early mobilisation, reduced blood loss, and a shorter hospital stay.<sup>2,3</sup> However, there are a number of potential drawbacks. MICS has been associated with additional complications to those after sternotomy, not all patient groups are suitable, and the surgical skills required to master video-assisted mitral valve repair procedures may involve a substantial learning curve to achieve expertise.

## MIDCAB surgery

MIDCAB surgery describes a minimally invasive approach to coronary artery bypass grafting. This is usually performed using an ‘off-pump’ technique, avoiding the use of CPB and cardioplegia solution. Access is via a left anterior mini-thoracotomy plus additional ports, leading to faster recovery times and a shorter intensive care stay.<sup>4,7</sup> However, it is only suitable for one or two vessel coronary grafting, and careful patient selection is essential.

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**Table 1** Summary of differences between two different MICS approaches

|                                 | Right anterolateral VATS/<br>minithoracotomy                                                                                                                            | Minimal access aortic<br>valve replacement              |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Lesions appropriate             | Mitral regurgitation<br>Tricuspid Regurgitation<br>Myxoma<br>ASD/PFO<br>Atrial fibrillation                                                                             | Aortic stenosis                                         |
| Procedures possible             | MV repair or replacement<br>TV repair or replacement<br>Resection of myxoma<br>(L & R atria)<br>ASD/PFO closure<br>AF ablation + LAA clip (bilateral<br>VATS procedure) | AV replacement or repair<br>Ascending aorta replacement |
| Cardiopulmonary bypass cannulae | Femoral venous and arterial                                                                                                                                             | Femoral or Direct                                       |
| Aortic cross clamping technique | Endoclamp balloon via femoral arterial cannula<br>Direct aortic cross clamp (Chitwood)                                                                                  | Direct aortic cross clamp                               |

MV, mitral valve; TV, tricuspid valve; AV, aortic valve; L, left; R, right; ASD, atrial septal defect; PFO, patent foramen ovale; AF, atrial fibrillation; LAA, left atrial appendage.

## Robotic cardiac surgery

Robot-assisted cardiac surgery first began in 1997 using a simple voiced-activated robot. Current third-generation da Vinci<sup>®</sup> surgical robots provide high-resolution three-dimensional (3D) imaging, movement scaling, magnification of the surgical field by up to 10-fold, and the option of dual consoles for facilitation of surgical training. Despite the significant startup costs involved (\$1.5 million to purchase each system), robotic cardiac surgery is now well established with more than 1700 cases performed annually in the USA.<sup>5–7</sup>

## AF ablation surgery

Surgical approaches to AF ablation have developed substantially in the past decade. Long within the cardiologist's domain, catheter ablation by pulmonary vein isolation alone has been found to have inferior success rates for conversion to sinus rhythm than a surgical approach. There are many ways to achieve surgical AF ablation. Bilateral video-assisted thoracoscopic surgery (VATS) incisions, with or without the support of CPB, are possible, using mainly bipolar radiofrequency techniques, or a hybrid approach using both surgical and percutaneous endocardial catheter ablation by both surgeon and cardiologists in the same procedure. With a surgical approach, rates of sinus rhythm restoration at 12 months can be as high as 93%, with low complication rates and mortality.<sup>8</sup> The scope of this topic alone warrants its own review article.

## Minimally invasive VATS cardiac surgery

This describes cardiac surgery performed via a right anterolateral VATS (approx. <4 cm) incision or minithoracotomy (approx. >4 cm) incision plus accessory ports, using video-assisted ('keyhole') camera technology; it was first reported by Carpenter and colleagues in 1996.<sup>9</sup> The lesions amenable to this approach are listed in Table 1. Mitral regurgitation requiring MV repair surgery is by far the most common lesion encountered and accounts for a large proportion of the published patient data. A commonly used term for this procedure is minimally invasive mitral valve surgery (MIMVS), but the technique does encompass all the procedures in Table 1. We shall henceforth refer to this technique as MIMVS, for clarity.

Access to the heart and major vessels is via an incision at the fourth or fifth intercostal space, or transareolar incision around the nipple, with lung isolation allowing collapse of the right lung. These cases have the most differences to a sternotomy case for the anaesthetist, and hence, anaesthesia for MIMVS warrants a thorough explanation.

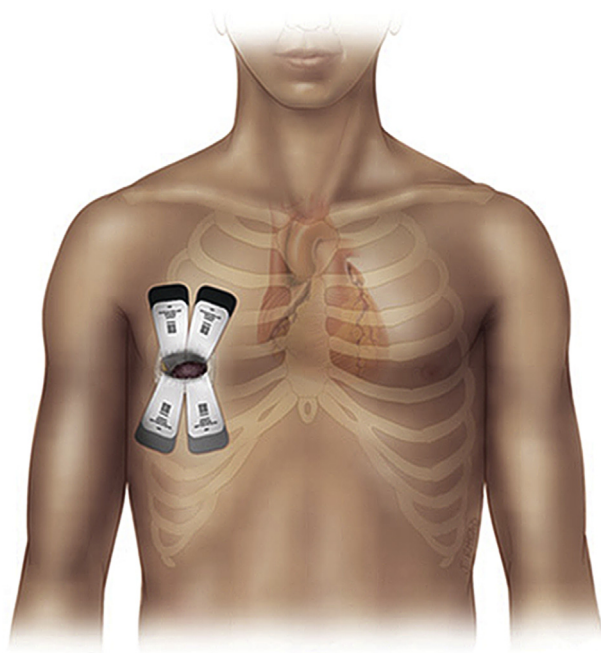
MIMVS cases encompass a number of features not usually encountered in cardiac surgery. Therefore, effective team working between the perfusionist, anaesthetist, surgeon, and operating theatre staff is essential. For the anaesthetist, a number of specific steps differ from a conventional sternotomy approach. These include the necessity of one-lung ventilation (OLV), use of cerebral oximetry, additional large-bore jugular vascular access, and advanced skills in TOE.

For the perfusionist, there are many differences. Peripheral cannulation for CPB is inherently more dangerous than direct cannulation of the aorta and vena cavae. A vacuum is used to assist venous drainage, which can cause air locks or haemolysis. Effective arrest of the heart is dependent on achieving good aortic occlusion with the aortic clamping method chosen. If an endoballoon clamping method is being used to deliver cardioplegia, the process of inflation of the balloon and delivery of cardioplegia is technically more demanding for the perfusionist. Also, the use of CO<sub>2</sub> to flood the surgical field throughout the procedure is mandatory, as deairing of the heart is harder to achieve at the end of surgery.

Overall, MIMVS procedures are safe and effective, with no significant difference found in mortality between MIMVS and conventional approaches for mitral surgery.<sup>3</sup> The locations on the chest wall of some of the surgical incision sites are shown in Figures 1–3. The main differences in approaches to the mitral valve area and aortic valve are also summarised in Table 1.

## Patient selection

Patient selection for MIMVS is essential. Elderly, frail patients may benefit from the less invasive approach. Aortic atherosclerotic disease and potential length of cross-clamp time may preclude the patient with multiple comorbidities. The cosmetically beneficial results are especially appealing to younger patients. MIMVS is ideally suited to an enhanced recovery approach to patient care. Most patients are expected to



**Fig 1** Right anterolateral incision for MIMVS and video-assisted cardiac surgery. Reproduced with permission from Edwards Lifesciences.

recover faster and leave hospital earlier than after a conventional midline sternotomy.<sup>1</sup> It is possible to discharge younger patients home on the second postoperative day.

## Anaesthesia for MIMVS

### Preoperative assessment

History, examination, and investigations are reviewed. Preoperative investigations should be tailored to each patient individually but usually include a similar workup to those undergoing sternotomy. These include routine blood tests, ECG, transthoracic echocardiogram (TTE), transoesophageal

echocardiogram (TOE), or TTE and TOE, coronary angiography, and pulmonary function tests. Any factors are identified which may make a minimally invasive approach impractical or impossible. A computed tomography (CT) scan of the arterial tree assesses the calibre of femoral arteries and aortic dimensions, and determines the degree of any aortic atheromatous disease. Absolute and relative contraindications to the technique are detailed in [Table 2](#).

The pros and cons of premedication should be considered. An 'enhanced recovery' approach avoids sedative premedication for a quicker recovery. Most patients will take a number of cardiac medications, which are managed similarly as for conventional sternotomy procedures.

### Monitoring

Standard cardiac anaesthesia monitoring is established. Depth of anaesthesia monitoring is useful. Cerebral oximetry should also be considered. Cerebral oximetry uses near-infrared spectroscopy to monitor cerebral venous oxygenation during the procedure via electrodes placed on the forehead. Baselines are set whilst breathing room air, which are typically 60–75%. It allows identification of reduced perfusion to the brain or whole body. Unilateral cerebral hypoperfusion can occur, if an endoballoon device migrates distally and occludes the right brachiocephalic artery. When selective antegrade cerebral perfusion is being instituted via the right subclavian artery, usually for aortic arch surgery, cerebral oximetry can detect the rare instance of an aberrant right subclavian (Lusorian) artery. Bilateral cerebral venous desaturation is a surrogate of whole body hypoperfusion, as seen when flows are decreased before surgical manipulations. Cerebral oximetry is discussed further in the section on complications.

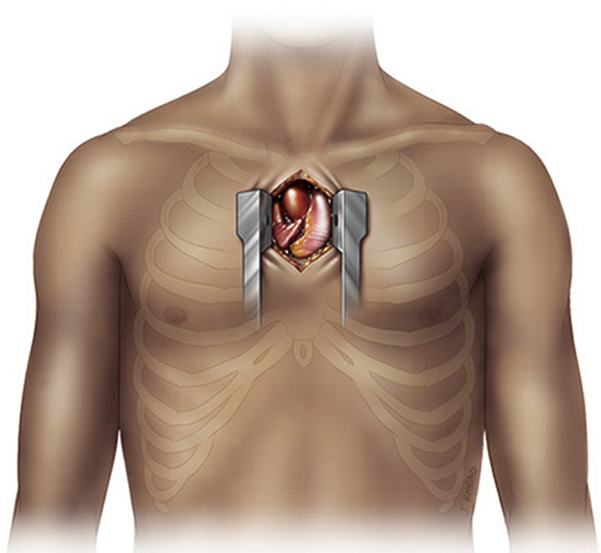
External defibrillation pads should always be placed on the chest for MIMVS and minimally invasive aortic procedures, as use of internal defibrillation paddles is not possible. Occasionally, paediatric-sized paddles may fit around the heart in a ministernotomy approach. The empty right lung field in a MIMVS case will cause increased transthoracic impedance, impairing current flow. Therefore, temporary reinflation of the right lung is necessary if defibrillation is required.

If an intra-aortic endoballoon approach is to be used for aortic clamping, then bilateral radial arterial cannulae will be necessary to monitor migration of the balloon distally to the brachiocephalic trunk/innominate artery (see [Figs 4 and 5](#)). Hence, invasive pressure monitoring should include four transducers to encompass two arterial lines, CVP and a transducer for the tip of the endoballoon. The anaesthetic monitor display ideally has overlaid waveforms of all of these.

### Induction and maintenance

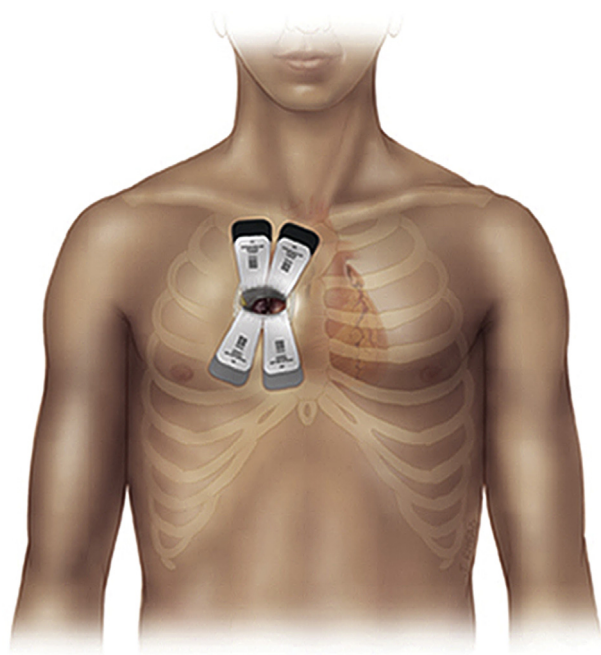
After preoxygenation, anaesthesia is typically induced using a combination of short-acting opioid and propofol, followed by neuromuscular block. To facilitate OLV, a double-lumen tracheal tube is usually placed and checked as for any standard thoracic lung isolation case. In the context of a difficult airway, a single-lumen tube with a bronchial blocker is an adequate alternative. Some centres use a single-lumen tube and commence CPB before opening the chest.

Anaesthesia is then maintained using either a volatile agent or propofol infusion. Monitoring of core temperature using a bladder temperature catheter in addition to a



**Fig 2** Mini-sternotomy incision. Reproduced with permission from Edwards Lifesciences.



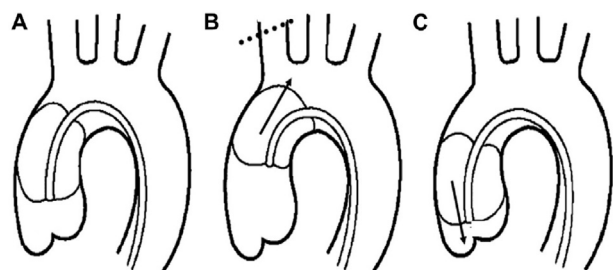


**Fig 3** Right anterior thoracotomy (RAT) incision. Reproduced with permission from Edwards Lifesciences.

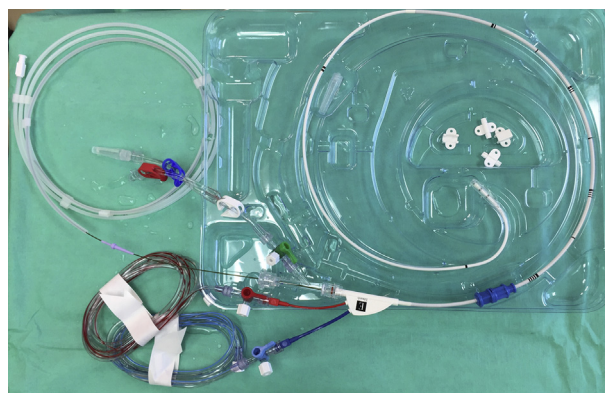
peripheral nasopharyngeal temperature probe is important. Hypothermia during CPB may be required, which allows flow rates to be decreased and an assumed degree of extra cerebral protection. Cooling of the patient is commenced if high pressures result from the blood flow rate in the femoral arterial cannula, or if the cerebral oximetry suggests poor cerebral perfusion. Forced air warmers or underbody warming blankets plus fluid heating systems should be used to maintain normothermia after separation from bypass. Routine antibiotic prophylaxis should be administered before skin incision.

### Transoesophageal echocardiography

The role of TOE in cardiac surgery is now well established with evidence showing it can reduce cardiovascular complications.<sup>10</sup> MIMVS cannot be performed without TOE. The pre-operative investigations, including CT and TOE, should have already highlighted contraindications or further pathology. In addition, a further comprehensive perioperative TOE is performed in the anaesthetic room, before insertion of internal



**Fig 4** Positioning of the endoballoon within the ascending aorta: (A) Ideal position. (B) Distal migration may occlude the innominate artery (dotted line). (C) Proximal migration may impede delivery of cardioplegia to the coronary arteries. Reproduced with permission from Wiley.



**Fig 5** The Edwards Lifesciences IntraClude II endoballoon device.

jugular vein catheters. Lesions that may prevent proceeding with MIMVS include aortic dilatation or significant regurgitation (Table 2). Additional pathology may require opening of the right heart, in which case the anaesthetist inserts a superior vena cava (SVC) cannula (see vascular access section).

If the mitral valve is being repaired, then a full assessment of the mitral valve is performed. The mechanism of mitral regurgitation is elucidated, if not already known, including which leaflets are involved. Mitral annular measurements, the anterior leaflet length, and intertrigonal distance will assist in the surgeon's decision of which size annuloplasty ring to insert. The risk of developing systolic anterior motion (SAM) of the mitral anterior leaflet after repair is also assessed. Left and right ventricular function plus pulmonary artery pressures are measured, to aid in the decision of whether a pulmonary artery catheter is required to guide weaning of inotropic support after the operation. Use of an endoballoon, rather than a direct aortic clamp, is generally not advised if the ascending aorta exceeds 40 mm.<sup>11</sup>

The ascending and descending aorta are interrogated for atheromatous plaques, which are a contraindication to MIMVS. Destabilisation of a plaque during threading of the wire or placement of the endoballoon could lead to plaque disruption and systemic embolisation. The aortic valve is assessed for competency and the presence of a distinct sinotubular junction, which assists correct positioning of the endoballoon. Demonstration of normal coronary artery anatomy is reassuring, as aberrant anatomy may compromise flow of cardioplegia when using the endoballoon aortic clamp (Fig. 6 and Fig. 6 online video). 3D images of the prolapsing or flail segments of a regurgitant mitral valve are potentially helpful to the surgeon but are not mandatory.

### Vascular access

The usual method of venous cannulation for left heart procedures is a long, two-stage SVC/inferior vena cava (IVC) cannula, inserted via the femoral vein. However, if the right side of the heart is to be opened to the atmosphere, planned or inadvertently, then air can be entrained into the bypass circuitry, potentially causing a dangerous air lock for the perfusionist. Surgery involving the right atrium would necessitate this, for example tricuspid valve surgery, atrial myxoma, atrial septal defect (ASD), or patent foramen ovale (PFO) repair. In this case, SVC cannulation is required, so that the equivalent of bicaval cannulation is provided for the perfusionist. The

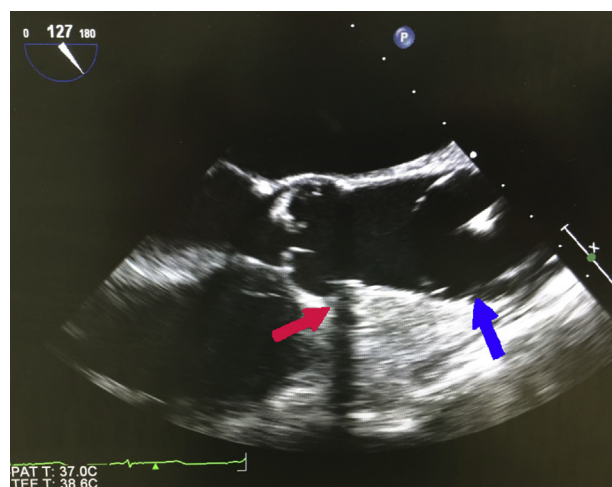
**Table 2** Absolute and relative contraindications to MIMVS

| Absolute contraindications                              | Relative contraindications                |
|---------------------------------------------------------|-------------------------------------------|
| Aneurysm of ascending aorta >40 mm                      | Pectus excavatum chest deformities        |
| No defined sinotubular junction                         | Pleural adhesions                         |
| Aortic regurgitation >moderate                          | History of chest trauma/rib fractures     |
| Inability to use TOE (oesophageal stricture, achalasia) | Previous radiotherapy                     |
| Mobile aortic atheroma                                  | Redo surgery (but potentially beneficial) |
|                                                         | Morbid obesity                            |
|                                                         | Aortic atheromatous plaques               |
|                                                         | Extreme mitral annular calcification      |

right internal jugular vein is preferred for an SVC cannula, as it has a near-linear path to the right atrium. The anaesthetist performs the insertion of this 17F or 14F line during routine central venous catheter insertion. However, the procedure is not without potential complications: those of any large bore cannula. Heparin 10,000 IU i.v. is given before insertion, with placement of the tip at the right atrial/SVC junction confirmed by TOE. Some centres use a retrograde cardioplegia coronary sinus catheter, which is sited by the anaesthetist via the right internal jugular vein using TOE guidance also. Maintaining sterility, the tubing is later connected through the surgical drapes via a Y-connector to the IVC cannula CPB limb.

### Conduct of surgery

The surgeon should position the patient carefully, with a gel roll or inflatable bag placed under the right hemithorax to open the rib spaces anteriorly. The right arm is carefully padded and secured down and away from the chest to allow access for the surgeon and camera arm. Care should be taken to avoid excessive traction on the brachial plexus. Padding of any pressure points at the elbows, wrists, and heels is ensured.



**Fig 6** Endoballoon (blue arrow) aortic occlusion device inflated in the ascending aorta in the midoesophageal aortic valve long axis view on TOE. The right coronary artery (red arrow) can be seen when confirming normal coronary anatomy. If reading online, click on the image to view the video.

In order to establish CPB, femoral venous cannulation is always performed first, guided by the TOE bicaval view (see Fig. 9 online-only video). This is in contrast to full sternotomy, where the aortic cannula is always inserted first. Full heparinisation must be achieved before CPB cannulae insertion. The surgeons may have scanned the femoral vessels using ultrasound before commencing surgery, in addition to pre-operative CT measurements, to assess the patency and calibre of the femoral vessels. Some patient groups have narrow femoral arteries, particularly female smokers, which may impact on the pressures generated in the arterial cannula at full flow. The risks include arterial dissection and haemolysis. If the pressure in the arterial cannula is >300 mm Hg, then the endoballoon, if being used, should be inserted in the opposite groin to the arterial CPB cannula. Once CPB is established, the descending aorta is visualised with TOE for several minutes to rule out aortic dissection.

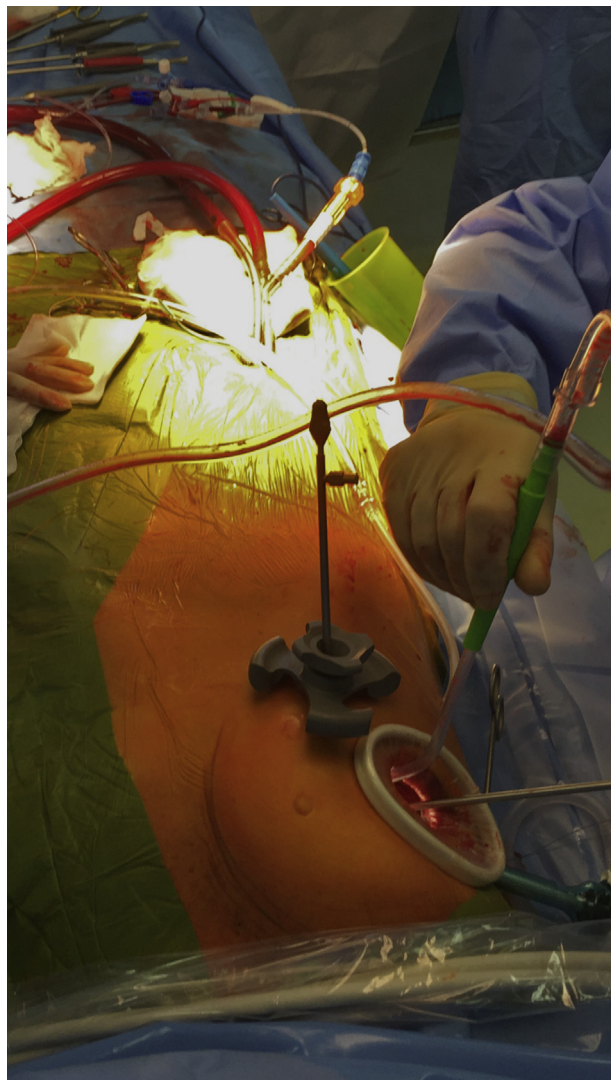
Before diastolic cardiac arrest with cardioplegia, the ascending aorta is clamped. There are two main methods: application of a transthoracic (Chitwood, Scanlan International Inc., Minneapolis, MN) clamp directly across the aorta or inflation of an endoballoon inserted into the ascending aorta via a side-arm on the arterial cannula. The Chitwood clamp may be used safely up to an ascending aorta diameter of 45 mm. Using a Chitwood clamp necessitates a cardioplegia cannula in the ascending aorta, below the clamp, also through the chest wall. The endoballoon, however, allows delivery of cardioplegia down the central port once the balloon is inflated (Fig. 7).

Correct placement of an endoballoon is one of the technically demanding points of the procedure, for the surgeon, perfusionist, and anaesthetist; hence, good communication is paramount. In the surgeon's hands, the sensation of the inflating balloon has been likened to 'flying a kite', with the pressure of the blood from the arterial CPB cannula opposing the force of blood being ejected from the heart. The perfusionist manipulates the systemic vascular resistance with a vasopressor to overcome the pressure generated by the ejecting heart whilst the endoballoon is being inflated, but not so much that it forces the balloon proximally towards the aortic valve. The anaesthetist constantly sees the ascending aorta in the TOE midoesophageal aortic valve long-axis view. A sudden unilateral pressure decrease in the right-sided arterial line trace on the monitors may represent distal migration of the balloon occluding the innominate artery, a complication thought to occur in approximately 7% of cases.<sup>12</sup>

Good teamwork allows that anyone in the operating team can alert the loss of the right radial artery trace, in this circumstance. Proximal migration of the endoballoon could lead to reduced delivery of cardioplegia to the coronary vessels and inadequate myocardial protection, whereas distal migration would compromise perfusion of the head and upper limbs. Cerebral oximetry monitoring can be very useful in alerting compromise to cerebral blood flow if migration occurs, but also as a surrogate marker for whole body perfusion. In contrast, the transthoracic Chitwood clamp appears to be cheaper and quicker, and result in fewer complications.<sup>1</sup>

At near-maximal inflation, the walls of the aorta begin to indent with the balloon. At this point adenosine is injected into the aortic root via the endoballoon central port. During the brief resulting asystole, cardioplegia is commenced (see Fig. 8 and Fig. 8 online video). TOE visualisation of the cardioplegia running down the left and right coronary arteries is reassuring. Ventilation is ceased when full CPB is achieved





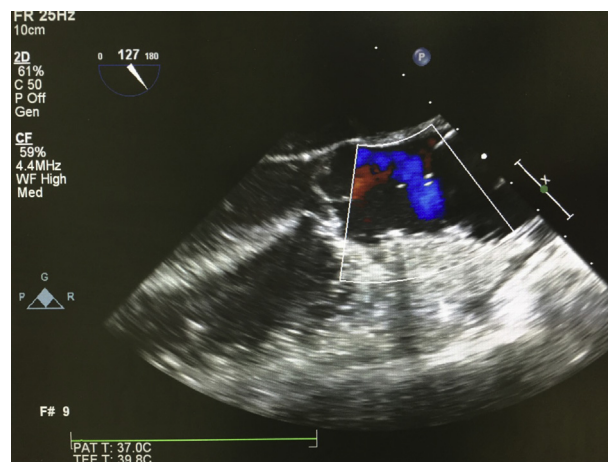
**Fig 7** The anaesthetist's view of a port access mitral valve repair case. The camera passes into the thoracic cavity through a small port lateral to the right anterolateral thoracic incision. The femoral vessel CPB cannulae can be seen, with the white endoballoon in place, passing through a side branch of the arterial cannula.

and anaesthetic vapour delivered through the CPB circuit or intravenously.

The left atrium is opened, and surgery is performed. Once the left atrium is open, TOE visualisation of the ascending aorta in the midesophageal views is lost. In select patients a deep transgastric view may allow confirmation that the balloon has not moved. In place of TOE images at this stage, bilateral radial arterial pressures are monitored.

Video-assisted 3D technology may be used to perform a mitral repair. When the procedure is completed, TOE assesses deairing of the chambers and the adequacy of the valve repair before separating from CPB (see Figs. 10 & 11 online-only videos). Left and right ventricular function can also be assessed, along with detection of immediate complications such as SAM of the anterior mitral leaflet.

As with conventional cardiac surgery, a temporary right atrial or ventricular epicardial pacing wire may be placed at the end of the procedure, before the pericardium is closed,



**Fig 8** TOE midesophageal aortic valve long axis view. Cardioplegia solution has been delivered to the coronary arteries via the central lumen of the endoballoon. Asystolic cardiac arrest is shown on the green ECG trace at the bottom of the image. If reading online, click on the image to view the video.

depending on the rate and rhythm after operation and after bypass. These wires may be removed in the same fashion as with a sternotomy procedure.

Intraoperative analgesia comprises a combination of short-acting synthetic opioid, morphine, and paracetamol. Infiltration of local anaesthetic, intercostal nerve blocks, or regional block are beneficial. Ultrasound-guided paravertebral block and catheter allow a paravertebral infusion after operation.

### Postoperative care

Selected patients may be suitable for tracheal extubation on table at the end of the procedure, or may be transferred intubated to a critical care area, to be extubated as soon as feasible. If a double-lumen tube has been used, it should be exchanged for a single-lumen tube before the patient leaves the operating theatre. Usual criteria for tracheal extubation apply: normothermia, haemodynamic stability, without significant acidbase disturbance, and appropriate analgesia instituted. The most consistent finding in several studies is a reduction in pain and faster return to normal activity. Patients who have minimally invasive surgery as a second procedure state that recovery was more rapid and less painful than their original sternotomy.

### Complications

A recent consensus statement, based on the limited data available, has suggested comparable short-term and long-term mortality, in-hospital morbidity (renal, pulmonary, cardiac complications, pain perception, and readmissions) and reduced sternal complications, transfusions, postoperative AF, duration of ventilation, and ICU and hospital length of stay.<sup>3,13</sup> However, the complications below are specific to this procedure.

### Conversion to sternotomy

Intraoperative conversion to sternotomy may be required, either if TOE findings are unfavourable or to manage intraoperative complications. One study reported that 1% of 3125

MIMVS cases were converted to sternotomy, the main reason being bleeding. The 30-day mortality then exceeded 23% in this patient group.<sup>14</sup> Reasons to convert include bleeding, pulmonary adhesions, aortic dissection, or poor exposure of the mitral valve.

### Postoperative bleeding

After operation, assessment of bleeding can be complicated by an incompletely opened pericardium, difficulty in viewing posterior collections of blood with transthoracic echocardiography, and 'hidden' bleeding into the right hemithorax. There should be a low threshold for repeat TOE. Reopening for bleeding may be successfully addressed by a repeat VATS procedure, as the culprit is often an intercostal vessel.

### Peripheral cannulation for CPB

Problems associated with peripheral cannulation for CPB are cited as drawbacks to MICS. Increased risk of neurological complications, stroke, and aortic dissection appear to relate to the use of retrograde arterial perfusion for CPB.<sup>15,16</sup> The risk of stroke is 2.1% vs 1.2% for sternotomy and the risk of aortic dissection is 2% vs 0%.<sup>3</sup> Patients with significant aortic atheroma are especially at risk, hence recommendations for preoperative imaging of potential vascular atheroma.<sup>17</sup> In addition, femoral compartment syndrome and femoral arterial pseudoaneurysm have also been reported, with some institutions describing an incidence of 1–6%.<sup>18</sup>

### Nerve injury

The ideal position for the patient on the operating table involves a slight lowering of the right shoulder. This can result in excessive retraction and injury to the brachial plexus. Opening the pericardium too posteriorly may lead to phrenic nerve damage, which has a risk of 3% compared with 0% for open surgery.<sup>3</sup> Placing the pericardiotomy 3 cm anterior to the nerve may mitigate this risk.<sup>1</sup> As with any prolonged anaesthetic, protecting pressure points is essential to avoid nerve compression injuries, particularly the ulnar nerve at the elbow.

### Harlequin syndrome

One of the benefits of using cerebral oximetry in port access cases is to detect differential cerebral perfusion. If the heart does not empty fully when established on CPB, potentially because of suboptimal venous drainage, the heart may still be ejecting deoxygenated blood. At this point, it is possible for the pressure of the deoxygenated blood reaching the right side of the brain to equal the pressure of the retrograde arterial oxygenated blood, as it reaches the left side of the brain. Consequently, oxygenation of the right cerebral hemisphere may be poor, potentially resulting in hemispherical ischaemia. This is known as the harlequin syndrome. Continuing to ventilate the left lung with 100% oxygen after initiation of CPB until diastolic cardiac arrest should prevent this potentially catastrophic complication.

### Complications of OLV

These complications are the same as those for any procedure requiring OLV and include acute lung injury (ALI), air leaks from inadvertent instrumentation of the lung, and segmental

or lobar collapse. A poorly recognised complication of OLV, after MIMVS, is re-expansion pulmonary oedema.<sup>19</sup> One retrospective study published an incidence of 2.1%, with a 12-fold increase in the 30-day mortality. The use of preoperative steroids or immunosuppressants and a prolonged aortic cross-clamp time ( $\geq 156$  min) were found to be independent risk factors. A lung-protective strategy, similar to any OLV procedure, must be adopted.

### Minimal access aortic valve replacement

Aortic valve surgery in the form of minimal access aortic valve replacement (mAVR) involves a small chest wall incision as opposed to full sternotomy, to reduce the invasiveness of the procedure. Common approaches are ministernotomy and right anterior minithoracotomy (RAT) (see [Figs 2 and 3](#)). Both approaches may utilise central or femoral vascular access for CPB cannulation. Published reports suggest shorter ventilation time, reduced postoperative pain, bleeding and incidence of transfusion, fewer wound infections, shorter intensive care stay, and a quicker return to normal activities.<sup>20</sup> The presence of an undisturbed distal pericardium with few adhesions theoretically improves the operating conditions should further cardiac surgery be necessary. A Cochrane review in 2017 found no difference in mortality, bypass time, aortic clamp time, or the incidence of major adverse events between mAVR and conventional sternotomy. There was, however, a small reduction in postoperative blood loss and length of cardiac surgery ICU (CICU) stay in the mAVR group.<sup>21</sup> Patients who are high risk surgically, but not suitable for transcatheter aortic valve implantation (TAVI), may benefit from a sutureless or rapid deployment mAVR procedure.<sup>22</sup>

### Anaesthetic considerations

The main principles of anaesthesia for routine full sternotomy apply to a ministernotomy approach, but the limited nature of the incision for mAVR does present some specific challenges of which the anaesthetist must be aware. The surgical incision is typically about one third of the normal sternotomy length: usually the sternum is opened from sternal notch to the second or third intercostal space. This not only limits surgical access but also the anaesthetist's view of the myocardium, especially the right ventricle (RV). The use of TOE is of additional benefit to monitor contractility in the absence of a full direct view of the heart. The intact lower sternum may help decrease pain levels and assist in pulmonary function after operation.

If femoral access for CPB is being used, TOE is essential to confirm placement of wires in the correct vessels. TOE is also necessary for confirming the position of any intracardiac venting catheters, such as placement of a catheter into the right upper pulmonary vein to gain access to the left ventricle. This is used to keep the heart empty during the procedure and assist in deairing of the heart before the aortic cross-clamp being removed.

Should cardioversion or defibrillation be required during surgery, it may be impossible to fit internal paddles through the ministernotomy, although it is sometimes feasible to use paediatric-sized paddles. Hence, external defibrillation pads are placed before induction. As with VATS cardiac surgery, the partial sternotomy incision hinders the ability to rapidly reopen the chest in the setting of a postoperative cardiac

arrest, with the consequential delay in delivering internal cardiac massage.

Conversion to full sternotomy is a significant setback for both the patient and surgical team, because it is associated with an increased risk of bleeding and subsequent prolonged critical care admission. In one study, aortic cross clamp and CPB duration was found to double after conversion to full sternotomy compared with conventional aortic valve replacement performed with elective full sternotomy.<sup>21</sup> The future direction of mAVR is developing rapidly, but further studies are required to reveal which of the many techniques are safest and best for the patient.

## Declaration of interest

The authors declare that they have no conflicts of interest.

## MCQs

The associated MCQs (to support CME/CPD activity) will be accessible at [www.bjaed.org/cme/home](http://www.bjaed.org/cme/home) by subscribers to BJA Education.

## Supplementary video

Supplementary video related to this article can be found at <https://doi.org/10.1016/j.bjae.2018.06.004>.

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